

Stephanie Buttigieg

PHD STUDENT · INSTITUTE OF ASTRONOMY, UNIVERSITY OF CAMBRIDGE

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Research interests

- Supermassive black holes across cosmic time
- Cosmological simulations
- Gravitational waves

Education

Institute of Astronomy, University of Cambridge

Cambridge, UK

PHD IN ASTRONOMY

October 2023 – present

- Supervisors: Prof. Debora Sijacki and Dr Christopher J. Moore
- Research area: Supermassive black hole mergers in cosmological simulations
- Core member, LISA Consortium

University of Cambridge

Cambridge, UK

MAST IN ASTROPHYSICS

October 2022 – June 2023

- Graduated with distinction
- Final Project: *Is space haunted? Exorcising ghosts from the gravitational particle spectrum*

University of Malta

Msida, Malta

BSC IN MATHEMATICS AND PHYSICS

October 2018 – September 2022

- Graduated with first class honours; ranked top of class
- Exchange year at the University of Warwick, UK (September 2020 - June 2021)
- Final Year Project: *Dynamical Systems in Beyond Standard Theory Cosmology*

Publications

PUBLISHED

Buttigieg, S., Sijacki, D., Moore, C. J., Bourne, M. A., 2025. *Premature supermassive black hole mergers in cosmological simulations of structure formation*. Published in the Monthly Notices of the Royal Astronomical Society 2019 (2025), 2038 <https://doi.org/10.1093/mnras/staf1336>. e-Print: [arXiv:2504.17549](https://arxiv.org/abs/2504.17549).

IN SUBMISSION

Buttigieg, S., Sijacki, D., Moore, C. J., Bourne, M. A., Sesana A., 2026. *Comparing gravitational wave background predictions from cosmological simulations to pulsar timing observations*. Submitted to the Monthly Notices of the Royal Astronomical Society. e-Print: [arXiv:2607.05208](https://arxiv.org/abs/2607.05208).

Cooke, G. J., White, S. B., **Buttigieg, S.**, et al., 2026. *Women's Pioneering Scientific Contributions Absent in UK-wide Education Specifications* Submitted to the British Educational Research Journal.

Presentations

INVITED TALKS

March 2025. CamGW Meeting, Kavli Institute for Cosmology, University of Cambridge

May 2025. NANOGrav Astrophysics Working Group (teleconference)

November 2025. Galaxy and SMBH formation, growth, and co-evolution workshop, Nepal.

January 2026. STScI Galaxy & AGN Journal Club (teleconference)

May 2026. University of Helsinki Astrophysics Seminar.

July 2026. Computational Galaxy Formation Workshop, Germany.

September 2026. Kavli Astrophysics Symposium 2026: Forging the Frontier of Cosmic Discovery, Cambridge, UK.

CONTRIBUTED TALKS

February 2025. Wednesday Seminar Series. Institute of Astronomy, University of Cambridge.

May 2025. Unveiling massive black hole evolution with gravitational waves and light workshop. Paris, France.

July 2025. GR – Amaldi conference. Glasgow, UK.

September 2025. Young astronomers on galactic nuclei meeting. Southampton, UK.

September 2025. Black holes across cosmic time conference. Cambridge, UK.

March 2026. CamGW Meeting, Kavli Institute for Cosmology, University of Cambridge.

March 2026. Emerging Research Seminar Series, Christ's College, University of Cambridge.

October 2026. Young astronomers on galactic nuclei meeting. Seville, Spain.

Awards, Fellowships, & Grants

2023 - 27	Cambridge Centre for Doctoral Training in Data-Intensive Science PhD Studentship , UK Science and Technology Facilities Council (STFC)	~ £20,000 p.a.
2026	Community Service Award , Institute of Astronomy, University of Cambridge	
2022 - 23	Tertiary Education Scholarship Scheme , Government of Malta	£10,000
2025	Education and Outreach Grant , Royal Astronomical Society, awarded as part of the IncludeHer UK collaboration	£5,000
2023	Field's Scholarship , Christ's College Cambridge	
2025, 2026	Bottomley Fund for Travel , Christ's College, Cambridge	
2023	Evolve Prize for Science , University of Malta	
2019, 20, 22	Dean's List , Faculty of Science, University of Malta	

Teaching Experience

2025	Mathematics for Natural Science Tripos Part Ia , Supervisor, Christ's College, University of Cambridge
2018 - 2024	Mathematics and Physics , Private Tutor
2019	PwC Malta , STEM Trainer

Professional & Industry Experience

Visa

RESEARCH INTERN

March 2026 - September 2026

LLM interpretability and debiasing via activation steering

European Organization for Nuclear Research (CERN)

SUMMER RESEARCH INTERN

June 2022 - August 2022

Data quality tests for the FLUKA software

Skills

PROGRAMMING

Python (NumPy, SciPy, Astropy, PyTorch, Matplotlib), Bash

COMPUTING

HPC cluster computing, GPU computing, Git

COSMOLOGICAL SIMULATIONS

Post-processing and analysis of simulation outputs

STATISTICAL & MACHINE LEARNING

Bayesian inference and MCMC, large language models interpretability

LANGUAGES

Maltese and English (native), Italian (basic conversational)

Service, Outreach & Leadership

SCIENCE COMMUNICATION & WOMEN IN SCIENCE ADVOCACY

Ongoing. Creator and presenter of *Astronomija bil-Malti*, an online science communication series in Maltese.

October 2024. Public speaker. Institute of Astronomy Open Evenings. Talk titled *Meet the Cosmic Giants: Understanding Supermassive Black Holes*.

February 2025. Invited panellist. *Curiosity Knows No Boundaries: Celebrating Women and Girls in Science*, organised by the Malta Chamber of Scientists at the University of Malta.

March 2025. Invited panellist. *Her Voice in Science*, European Parliament.

January 2026. Public speaker. Institute of Astronomy Open Evenings. Talk titled *Listening to the Universe with Gravitational Waves*.

PROFESSIONAL APPOINTMENTS

January 2025 – January 2026. Co-chair, Women's Network, Institute of Astronomy.

September 2025 – September 2026. Local Organising Committee member, *Black holes across cosmic time*.

March 2026 – September 2026. Local Organising Committee member, *Kavli Astrophysics Symposium 2026: Forging the Frontier of Cosmic Discovery*.

October 2024 – October 2025. PhD Academic Representative, Faculty Board for Physics and Chemistry, University of Cambridge.

October 2022 – March 2024. International Officer; Women's & Non-Binary Officer; Vice President, Christ's College MCR.

October 2021 – June 2022. Student Representative, Board of Studies, Department of Physics, University of Malta.

July 2019 – June 2020. Education Officer, Science Students' Society, University of Malta.